

IH-53 EV bill of materials

Purchase list for the wiring in [wires.md](#) / [IH53EV.kicad_sch](#). Lengths are **buy quantities** (spool / cut with slack), not measured harness lengths.

How to use: skip anything already on the truck. Factory pigtailed and their housings ship with the device — do not rebuy those halves. Buy the vehicle-side mate, contacts, and bulk wire.

Tesla modules **BT2–BT6** in series are ~108–126 V. HV insulation on this sheet is 600–1000 V; interrupting ratings must still be **DC** at pack voltage.

1. Major assemblies (confirm on hand)

Qty	Item	MPN / spec	Used for	Status
5	Tesla Model S battery modules	BT2–BT6	traction pack	on hand
1	12 V vehicle battery	BT1	Blue Sea / DCIS 12 V	on hand / buy if needed
1	NetGain HyPer 9 + K1 AMPSEAL harness	AMPSEAL 776164-1	inverter	on hand
1	Orion BMS 2 + CWHMIO 18 AWG Main I/O	TE 1376360-1, Sumitomo 8240-4892	BMS	on hand
1	Orion cell-tap harness	TE 1318389-1, 22 AWG	module taps (not on this sheet)	on hand / separate
1	Thunderstruck TSM2500	includes DJ7031-4.8 + SB50 pigtailed	charger	on hand
1	Fast and Quiet DCIS 600 W	includes dj70310-6.3-11 + DJ7021-8-21	pack → 12 V	on hand
1	Ewert CANdapter	DE-9 CAN, USB-B	CAN config	on hand

2. Buy — HV switching and protection

Qty	Item	MPN / spec	Used for	Where
1	Pack fuse 500 A, bolted semiconductor	Mersen A25X500-4 (500 A, 250 V)	F4, pack +	DigiKey / Mersen
1	Fuse block / pads for A25X (1.5" dia, bolt-in)	match A25X500-4	F4 mounting	with fuse

Qty	Item	MPN / spec	Used for	Where
1	Manual HV battery disconnect, 600 A, red handle, no aux	Rincon HVBD6AXR	SW1 kill switch	Rincon / EV distributors
1	HV contactor, 12 V coil, M8 power terminals	Gigavac / Sensata GV200QA-1	G1 pack +	Gigavac / EV distributors
1	Precharge / discharge relay, 12 V coil, SPST-NO, #250 QC	TE Potter & Brumfield T9AP1D52-12	K2	DigiKey / Mouser
1	Charger AC relay, 12 V coil, DPDT, #250 QC	TE Potter & Brumfield T92P11D22-12	Power_Relay1	DigiKey / Mouser
1	Pack-voltage key relay, 12 V coil, 400 V DC / 15 A, polarized contacts 3+ / 4-	Omron G9EJ-1-E-UVD DC12	K4 DCIS key	DigiKey / Mouser (not Amazon cube relays)
1	12 V continuous-duty solenoid, 200 A, insulated coil	Cole Hersee / Littelfuse 24213 (aka 24213-BX)	K3 switched 12 V	Waytek / NAPA
1	2-pole 1000 V DC breaker, 32 A	schematic 2Pole_1000V_DC_32Amp1 + Circuit_Breaker3 (both poles of one breaker)	TSM2500 DC +/-	PV/EV DC MCB (e.g. Noark Ex9BP 4P 1000 V class — confirm pole count vs 1000 V)
1	2-pole 1000 V DC breaker, 10 A	schematic 2Pole_1000V_DC_10Amp1 + Circuit_Breaker4	DCIS pack +/-	same family
1	1-pole 1000 V DC breaker, 10 A	schematic Circuit_Breaker2 / sheet text 1Pole_1000V_DC_10Amp	precharge	same family
1	MAXI fuse holder	Littelfuse 5006	F5	Waytek / DigiKey
1	MAXI fuse 50 A	MAXI 50 A	F5, 12 V Always_Hot	same
1	MAXI fuse holder	Littelfuse 5006	F6	same

Qty	Item	MPN / spec	Used for	Where
1	MAXI fuse 50 A	MAXI 50 A	F6, DCIS 12 V feed	same
1	5×20 mm fuse holder (panel or inline)	F5×20	F7	DigiKey
5	5×20 mm fuse 1 A , slow-blow (Recom T1A class)	schematic F5×20 (1A max)	F7 RAC02	DigiKey (spares)

Breaker Values are empty on the schematic; ratings above are from the reference designators and the **1Pole_1000V_DC_10Amp** note. Buy **DC-rated** PV/EV breakers, not AC-only DIN breakers.

3. Buy — 12 V, charge, CAN, inlet

Qty	Item	MPN / spec	Used for	Where
1	ST Blade split-bus fuse block	Blue Sea 5032	Always_Hot / Switched / fused circuits	Blue Sea / West Marine
1	ATO/ATC fuse assortment 5–15 A	—	Blue Sea circuits actually used (Always_On, Ready/Precharge, Key_Switch). Circuits 1–4 and 6 unused.	any
1	PCB AC/DC 2 W, 12 V out, 85–305 VAC	Recom RAC02-12SE/277 (schematic RAC02-12SE_277). If NRND, successor in RAC02-E/277 or RAC02-SK 12 V 2 W	charge 12 V for T92 / Orion CHARGE	DigiKey / Mouser
1	CAN isolator	Advantech BB-CANOP	HyPer CAN ↔ Orion CAN1	Advantech / DigiKey
1	SAE J1772 inlet (16–32 A)	any listed J1772 socket	AC + CP + PP (J1773 on the sheet)	EVSE suppliers
1	Ignition / key SPST	schematic SW3	K3 coil	automotive
1	Inertia / crash switch, SPST, NC or as wired	schematic SW4	series with ignition	automotive (GM-style inertia)
1	USB bulkhead or USB-A/B extension	USBConnector1	CANdapter USB	any

4. Buy — connectors (vehicle side)

Included pigtails: TSM2500 DJ7031 + SB50, DCIS dj70310 + DJ7021, Orion CWHMIO, HyPer K1. Buy mates / extra contacts only if you are building the vehicle half instead of splicing onto the pigtail.

Qty	Item	MPN / spec	Used for
1 pair	Anderson SB350 housings, 350 A / 600 V	Anderson SB350	pack / inverter disconnect
4	SB350 contacts for 2/0	Anderson 2/0 contacts (e.g. 905-series)	HV-09...HV-12
1	Anderson SB50 housing + 2× contacts for 6–10 AWG / 6 mm²	mate to TSM2500 DC pigtail	CH-DC+V / CH-DC–V
1	Weather Pack 4-way kit (both housings, TPA, cable seals)	Aptiv 4-way line 12015797 (tower) + 12010974 (shroud), or 4-way square 12015798 + 12015024 ; terminals 18 AWG (socket 12089188 / pin 12089040 class)	DelphiWeather4Pack1
1	Deutsch DTM 3 kit	DTM06-3S + DTM04-3P , wedges WM-3S / WM-3P, pins 1060-16-0122, sockets 1062-16-0122	inverter CAN bulkhead
1	DJ7031-4.8 vehicle half + 3× 4.8 mm contacts for 14 AWG	mate to TSM2500 AC	AC-CHG-L/N/PE
1	dj70310-6.3-11 vehicle half + contacts: 2× 14 AWG HV , 1× 18 AWG HV	mate to DCIS input/key	DC-IN+ (red), DC-IN– (black), DC-KEY (green)
1	DJ7021-8-21 vehicle half + 2× 8.0 mm contacts for 10 AWG	mate to DCIS 12 V	12-DJ+, 12-BAT–
—	Sumitomo 8240-4892	only if extending/re-pinning CWHMIO	Orion Main I/O

5. Buy — wire

Colors follow [wires.md](#). **HV is 600–1000 V silicone / EV**, not TXL. Traction 2/0 is **orange**; polarity is heatshrink, not jacket color.

Qty	Size / type	Color	Used for (wire IDs)
45 ft	2/0 DLO	orange	HV-01...HV-14 (all traction)
10 ft	10 AWG SGX (or GXL)	red	12-BAT+, 12-HOT, 12-SW
8 ft	10 AWG SGX	black	12-BAT–
10 ft	10 AWG HV	red	CH-DC+V, CH-DC+2
10 ft	10 AWG HV	black	CH-DC–V, CH-DC–2
10 ft	14 AWG HV	red	DC-IN+, DC-IN+2
10 ft	14 AWG HV	black	DC-IN–, DC-IN–2
8 ft	18 AWG HV	red	DC-K4TAP, PC-01, PC-02, PC-03
8 ft	18 AWG HV	green	DC-KEY
15 ft	14 AWG 600 V AC (THHN/MTW/SXL as preferred)	brown	AC-L1, AC-CHG-L
15 ft	14 AWG 600 V AC	blue	AC-L2, AC-CHG-N
15 ft	14 AWG 600 V AC	green-yellow	AC-GND
5 ft	18 AWG 300 VAC (or 18 AWG AC as labeled)	brown	AC-L1-F7, AC-L1-RAC
5 ft	18 AWG 300 VAC	blue	AC-L2-TAP
8 ft	10 AWG TXL	red	12-DJ+
25 ft	18 AWG TXL	red	12-IGN1, 12-IGN2, 12-KEY, leftover 12 V
15 ft	18 AWG TXL	black	12-K3GND, 12-K4C, CHG-12–
1 pack	2/0 adhesive heatshrink	red	+ ends of orange 2/0
1 pack	2/0 adhesive heatshrink	black	– ends of orange 2/0

Do **not** buy TXL for charger DC, DCIS pack input, DCIS key, or precharge. Do **not** buy yellow/gray 14 AWG HV for the vehicle DCIS pack runs (factory pigtail is already yellow/gray).

Orion CWHMIO already supplies Always_On (green), READY (blue), CHARGE (red), Charger_Safety (red/white), DISCHARGE_enable (blue/white), CP (orange/white), PP (brown/white), CAN1 pair, and typically ground. Extend those tails rather than replacing them.

HyPer K1 already supplies CAN (K1-13 / K1-2), precharge/key (K1-24), and coil (K1-25 / K1-26) — splice or pin those colors; do not guess.

6. Buy — lugs and terminals

Qty (min)	Item	Wire	Lands on
12	2/0 ring, 5/16"	2/0 DLO	TB1/2/3 5/16", module lugs, F4, SW1, SB350
4	2/0 ring, M8	2/0 DLO	Gigavac A1+ / A2-, HyPer B+ / B- if M8
6	10 AWG HV ring, 1/4"	10 AWG HV	TB1/2 1/4", charger DC breakers
6	14 AWG HV ring, 1/4"	14 AWG HV	TB1/2 1/4", DCIS breakers
2	18 AWG HV ring, 1/4"	18 AWG HV	TB3 1/4" precharge
8	10 AWG ring, 5/16"-24	10 AWG SGX	K3 studs, MAXI 5006, battery posts
4	10 AWG ring, #10-32	10 AWG SGX	Blue Sea Always_Hot / Switched / Battery-
8	18 AWG ring or spade, #8-32	18 AWG	Blue Sea circuits
6	18 AWG ring, #10 chassis	18 AWG TXL / HV	chassis GND
20	0.250" QC (insulated), 14 AWG	14 AWG AC	T92 pins 13/14/23/24
20	0.250" QC (insulated), 18 AWG	18 AWG	T92 A1/A2, T9A, G9EJ, Cole Hersee coil
10	18 AWG HV 0.250" QC	18 AWG HV	G9EJ contacts 3/4, T9A pins 3/4
1 pack	heat-shrink, adhesive lined, 10 AWG through 18 AWG	—	every small lug
1	3× HV junction / terminal block with 5/16" and 1/4" studs	TB1, TB2, TB3	make or buy copper bars

7. Optional / tooling

Qty	Item	Why
1	2/0 lug crimper (dies for 2/0 DLO)	traction cable
1	Weather Pack crimper + cavity tool	4-way bulkhead
1	Deutsch DTM (size 16) crimper	inverter CAN
1	14 / 18 AWG insulated-terminal crimper	QC and rings
—	HV cable loom, orange tape, warning labels	pack / charger DC / DCIS HV

8. Do not buy (included or unused)

Item	Why
TSM2500 AC DJ7031 pigtail, DC SB50 pigtail, CAN pair	ships with charger
DCIS dj70310 / DJ7021 pigtails	ships with DCIS
Orion CWHMIO 18 AWG Main I/O	ships with this BMS
HyPer 9 K1 AMPSEAL harness	ships with inverter
CANdapter USB-B cable	ships with adapter (add bulkhead only if you want a dash port)
Blue Sea circuits 1–4, 6	unused on this sheet
Orion fan pins 9 and 10	no-connect
Tesla cell taps on this schematic	separate Orion tap harness

9. Schematic vs this list

- Kill switch: **Rincon HVBD6AXR** (600 A, no aux, red).
- K4: **Omron G9EJ-1-E-UVD DC12** only — polarized HV contacts, 12 V coil.
- F5 and F6 are both **MAXI 50 A**; 12 V battery feeds are **10 AWG SGX**.
- Traction 2/0 is **orange** with red/black heatshrink at the lugs.
- Vehicle DCIS pack HV is **red / black** 14 AWG HV (factory pigtail stays yellow / gray).
- Precharge wire is unlabeled on the sheet; buy **18 AWG HV** red.
- RAC02 taps: buy **18 AWG 300 VAC** (schematic also says 18AWG AC).
- J1772 symbol reference is **J1773**.
- Circuit_Breaker3 is the second pole of the **32 A** charger DC breaker; Circuit_Breaker4 is the second pole of the **10 A** DCIS breaker.